

1

NUMBER AND ALGEBRA

INTEREST AND DEPRECIATION

The value of an investment increases over time as a result of interest being added to it, whether it be simple or compound interest. On the other hand, the value of assets and items such as cars and office equipment decreases over time due to age and wear-and-tear. Compound interest and depreciation use formulas that involve repeated percentage increase and decrease respectively.



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Chapter outline

		Working mathematically				
1.01	Earning an income	U	F	PS	R	
1.02	Income tax	U	F	PS		C
1.03	Simple interest	U	F	PS	R	
1.04	Compound interest	U	F			
1.05	The compound interest formula	U	F	PS	R	
1.06	Term payments	U	F	PS	R	C
1.07	Depreciation	U	F	PS	R	C

Wordbank

allowable deduction A part of a person's yearly income that is not taxed, such as work-related expenses and donations to charities

compound interest Interest calculated on the principal invested as well as on any accumulated interest

depreciation The decrease in the value of items over time due to ageing

future value The final amount of an investment after all interest has been added

net pay Pay received after deductions from gross pay; 'take-home' pay

per annum (p.a.) Per year

principal or **present value** The original amount of money invested or borrowed, for the purpose of calculating interest

repayment or **instalment** The amount of money paid at regular time periods (weekly, fortnightly, monthly) to pay off a loan

simple interest Interest calculated on the original principal invested only

Quiz
Wordbank 1

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (10):

1.01 Annual salaries • Overtime • Commission • Annual leave loading
1.02 Income tax
1.03 Simple interest • Simple interest rate
1.05 The compound interest formula • Compound interest
1.07 Depreciation

Twig video (1):

1.05 Could you owe more than America?

Quizzes (5):

- Wordbank 1
- SkillCheck 1
- Mental skills 1
- Language of maths 1
- Test yourself 1

Skillsheets (3):

SkillCheck Mental percentages • Percentage calculations
1.05 Spreadsheets

Worksheets (9):

SkillCheck Percentage shortcuts
1.01 Earning money
1.02 Income tax tables • Income tax and Medicare levy
1.03 Simple interest • Simple interest table
1.05 Compound interest
1.07 Depreciation
Mind map Interest and depreciation

Puzzles (4):

1.03 Simple interest
1.05 Compound interest with annual rests • Compound interest with non-annual rests
Language of maths Interest and depreciation crossword

Technology (3):

1.05 Comparing interest rates • Simple and compound interest calculator
1.06 Term payments

Spreadsheets (2):

1.05 Comparing interest rates • Simple and compound interest

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- calculate weekly, fortnightly, monthly and yearly incomes
- calculate wages, salaries, overtime, commission, piecework, royalties and annual leave loading
- use tables to calculate income tax and PAYG tax
- solve problems involving simple interest
- solve problems involving compound interest by repeated percentage increase
- solve problems involving the compound interest formula $FV = PV(1 + r)^n$
- solve problems involving term payments and short-term loans
- solve problems involving depreciation

SkillCheck ANSWERS ON P. 576

1 Convert each percentage to a decimal.

a 4% **b** 22% **c** 18.3% **d** 4.7%

e $9\frac{1}{2}\%$ **f** 6.75% **g** $15\frac{1}{4}\%$ **h** 20%

2 Find:

a 6% of \$1200 **b** 2.5% of \$4650 **c** 12% of \$37 450

3 Increase:

a \$7000 by 5% **b** \$3955 by 2% **c** \$8600 by 1.6%

4 How many months are there in:

a 3 years? **b** 2 years? **c** 5 years?

5 Copy and complete:

a One year = _____ weeks

b One year = _____ fortnights

c One year = _____ days

d 48 months = _____ years

e 84 days = _____ weeks

f 100 months = _____ years _____ months

6 If $P = mvt$, find:

a P when $m = 1600$, $v = 0.072$, $t = 10$

b m when $P = 120$, $v = 0.3$, $t = 8$

c v when $P = 18$, $m = 60$, $t = 5$

7 Evaluate, correct to the nearest cent:

a $\$5000 \times (1.045)^4$

b $\$28\,000 \times (1.03)^6$

c $\$15\,300 \times (1.065)^3$

d $\$32\,400 \times (1.072)^{10}$

Quiz SkillCheck 1

Skillsheets Mental percentages

Percentage calculations

Worksheet Percentage shortcuts

1

1.01 Earning an income

Wages, salaries and overtime

Videos
Annual salaries
Overtime
Commission

Worksheet
Earning money

A **wage** is calculated from the number of hours worked and is usually paid weekly. Wage earners can make more income by working extra hours (overtime).

A **salary** is a fixed annual amount, paid weekly, fortnightly or monthly. Salary earners do not earn overtime pay but can receive benefits such as a computer, company car, expense account, shares in the company or paid medical expenses.



Lyubov Levitskaya/Shutterstock.com

Units of time for wages and salaries

1 year = 12 months
1 fortnight = 2 weeks
1 year = 52 weeks for wage earners
1 year = 52.18 weeks for salary earners

The 2 most common rates of **overtime** pay are:

- **time-and-a-half** = 1.5 × normal hourly rate
- **double time** = 2 × normal hourly rate

Example 1

Tyrone earns a salary of $70 400 p.a. p.a. = per annum = per year
How much does he earn:

a each week? **b** each fortnight? **c** each month?

Solution

a Weekly income = $70 400 ÷ 52.18
= $1349.1759...
≈ $1349.18 Rounded to the nearest cent

b Fortnightly income = 2 × $1349.18 1 fortnight = 2 weeks
= $2698.36

c Monthly income = $70 400 ÷ 12 1 year = 12 months
= $5866.6666...
= $5866.67 Rounded to the nearest cent

Example 2

Noor earns $22.65 per hour at normal rates. Last week, she worked 38 hours at normal rates, 6 hours at time-and-a-half and 3 hours at double time. Calculate Noor's total earnings for the week.

Solution

Normal pay $= \$22.65 \times 38$

$= \$860.70$

Time-and-a-half pay $= 6 \times \$22.65 \times 1.5$ 6 hours

$= \$203.85$

Double time pay $= 3 \times \$22.65 \times 2$ 3 hours

$= \$135.90$

Total earnings $= \$860.70 + \$203.85 + \$135.90$

$= \$1200.45$

Commission, piecework and annual leave loading

Commission is earned by salespeople and agents, and is a percentage of the value of items sold or income made.

Piecework is earned according to the number of items made or tasks completed.

Royalties are paid to artists, authors and musicians for the sale or use of their work. When an artwork appears in a publication, a book is sold or a song or video is played to an audience, the artist or author earns a royalty, a percentage of the fee or price.



iStock.com/Allard1

Annual leave loading or **holiday loading** is extra pay given during annual leave (holidays) and is 17.5% of 4 weeks' normal pay.

Example 3

Georgia is a real estate agent and is paid a commission of 2.5% on the value of apartments she sells. She also receives a weekly retainer of $950. How much will Georgia earn if she sells an apartment for $590 000?

A **retainer** is a fixed amount paid regardless of how many items are sold.

Solution

Commission $=$ 2.5% of $590 000

$= \$14\,750$

Total earnings $=$ commission $+$ retainer

$= \$14\,750 + \$950$

$= \$15\,700$

$\therefore$ Georgia earns $15 700.

Example 4

Tobias is a songwriter who earns a royalty of 9.1 cents per stream for each song of length 5 minutes or less and an extra 1.75 cents per minute (or part thereof) for each song over 5 minutes. Last week, his songs 5 minutes or less were streamed 24 817 times and one song that is 5 minutes 35 seconds was streamed 4516 times. What were Tobias' royalties for the week, correct to the nearest cent?

'Part thereof' means that part of a minute is counted as a whole minute.

Solution

Royalty rate = 9.1 c

Royalty rate for 5 min 35 s song = 9.1 + 1.75 c

= 10.85 c

Tobias' royalties = $24\,817 \times \$0.091 + 4516 \times \$0.1085$ Express cents as dollars

= 2748.333

≈ \$2748.33



Stokkete/Dreamstime.com

Video
Annual leave loading

Example 5

Kham's annual salary is \$70 590. For his Christmas holidays, he received 4 weeks' normal pay plus 17.5% annual leave loading for the 4 weeks. Calculate Kham's:

a normal weekly pay

b annual leave loading

c total pay for the Christmas holiday.

Solution

a Weekly pay = \$70 590 ÷ 52.18

= \$1352.8171...

≈ \$1352.82

b Annual leave loading = $17.5\% \times \$1352.82 \times 4$ 17.5% of 4 weeks' pay

= \$946.974

≈ \$946.97

c Total holiday pay = $(4 \times \$1352.82) + \$946.97$ 4 weeks' pay + leave loading

= \$6358.25

EXERCISE 1.01 ANSWERS ON P. 576

Earning an income U F PS R

Express all answers correct to the nearest cent where necessary.

1 Find the weekly wage for each person.

a Sebeen earns $21.85 per hour and works for 40 hours.

b Connor works 8 hours a day, Monday to Friday, and is paid $23.47 per hour.

c Yoshe works on Monday and Tuesday from 8:30 a.m. until 4:00 p.m. and Thursday from midday until 9:00 p.m., and earns $30.60 per hour.

2 Greta earns $19.56 an hour and works for 31 hours each week. Chandler earns $21.44 per hour for his 27 hours of work. Who earns more per week and by how much?

3 Maggie earns a salary of $180 640 p.a. How much does she earn:

a each week? **b** each fortnight? **c** each month?

4 Rakitu considers 2 jobs, one locally with an annual salary of $57 640 p.a. and the other one in the city with a fortnightly pay of $2320. Calculate the weekly income for each job, determine which one pays more per week, and by how much. R

5 Anan works 38 hours at normal rates, 7 hours at time-and-a-half and 4 hours at double time. Calculate Anan's total earnings if he earns $19.60 per hour at normal rates.

6 Jacqui works 8.5 hours per day from Tuesday to Friday. She is paid $21.78 per hour. She also works on Saturday for 4.5 hours at a special rate of $24.59 per hour. How much did Jacqui earn for the week?

7 Idra works the following hours in a week at the clothing chain *Shop til U Drop*.

Day	Hours worked
Monday	9 a.m. – 5 p.m.
Tuesday	9 a.m. – 4 p.m.
Thursday	10 a.m. – 7:30 p.m.
Friday	10 a.m. – 5 p.m.
Saturday	10:30 a.m. – 5 p.m.

She is paid at the following rates.

Day	Rate of pay
Monday to Friday	$19.62 per hour
Saturday	$23.15 per hour
Thursday after 4:00 p.m.	

What is Idra's total income for the week? PS R

8 Fatimah is paid a commission of 2.5% on the value of goods she sells. She also receives a weekly retainer of $875. How much will Fatimah earn if she sells goods to the value of $41 600 in one week? Select the correct answer **A**, **B**, **C** or **D**.

A $1915 **B** $1061.88 **C** $2187.50 **D** $1018.13

EXAMPLE 1

EXAMPLE 2

EXAMPLE 3

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9 Nathan is a real estate agent whose commission is calculated on the value of the properties he sells:

- 3% paid on first $300 000
- 1.5% paid on next $250 000
- 0.75% paid on any value thereafter

How much commission did Nathan earn for selling a house for $625 000?

10 Briana designed an app, *KeyFinder*, that sells for $2.49. If she makes 70% profit on the sale price of each app sold, how much would she make from selling 800 units of this app?

11 Spriha is a university student and part-time rideshare driver. Last week, she drove for 16.5 hours and earned an average income of $33.15 per hour from passengers. If her job expenses are $16 per hour, how much did Spriha actually earn from her work last week?

12 Matt charges $60 for each lawn he mows and $45 for trimming hedges in each yard. In a week, he mows 24 lawns and trims 15 hedges. How much does he earn for the week?



Virrage Images/Shutterstock.com

13 *Clean 2 Swim* charges $86 to clean backyard pools. If this business earned $4644 in the first week of summer, how many pools were cleaned? R

14 Jade makes homemade eco-friendly soaps, shampoos and cleaning products. A customer purchases 3 homemade soaps, 2 bottles of shampoo and 3 of the cleaning sprays. How much does Jade receive for these purchases?



Homemade soaps $5.60



Eco-friendly shampoo $12.70



Natural cleaning spray $7.25

Volosina/Shutterstock.com, AlenKadr/Shutterstock.com, Lotus_studio/Shutterstock.com

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9780170479622

15 Maya is a songwriter who earns a royalty of 9.1 cents per stream for each song of length 5 minutes or less and an extra 1.75 cents per minute (or part thereof) for each song over 5 minutes. For one week, the number of times her songs were streamed are shown in this table.

EXAMPLE 4

Song length	Number of times streamed
5 min or less	15 406
Over 5 min up to 6 min	3216
Over 6 min up to 7 min	805

How much will Maya earn in royalties for the week, correct to the nearest cent?

16 A video content creator earns $9.20 per 1000 views on the online video platform *MeScreen*. If Dawson makes a video that is viewed 1 258 344 times in the first week, how much will he earn that week?

17 Tiago earns 19% in royalties on the sale of his novels. If 37 815 paperback copies of his novel are sold for $22.95 each and 16 432 e-books at $17.65 each, calculate his royalty payment.

18 Hasti developed a gaming app for preschoolers and earns 16% in royalties from the sales of the app, which is priced at $7.45 per download.

a Hasti's app was downloaded 7956 times last month. How much did she earn from those sales?

b How many apps must Hasti sell to earn $25 000? R

19 Calculate the annual leave loading for each person if it is 17.5% of 4 weeks' pay.

EXAMPLE 5

a Richard earns $1220 per week

b Ivy earns $2000 per fortnight

c Samir earns $5944 per month

d Ellie earns $46 630 p.a.

20 For his annual holidays, Jake received 4 weeks' normal pay plus 17.5% annual leave loading for the 4 weeks. If Jake's annual salary is $50 725, find his:

a normal weekly pay

b annual leave loading

c total pay for the 4-week holiday.

Did you know?

Residual royalties from old TV shows

Successful TV shows make money for production companies, and usually the writers and actors of these shows receive a share of the profits. They also receive residual royalties if these shows are given a second life through reruns and streaming on a different platform. In 2023, writers and actors in the US went on strike for five months to improve working conditions and achieve fairer distribution of profits received from streaming services.

Old TV series like *Friends* (1994–2004) and *The Office* (2005–2013) are still popular today, with a new generation of younger viewers often binge-watching them. The main cast of *Friends* still each earn 2% of the $1 billion (US) in revenue generated from this series each year, meaning they each receive royalties of $20 million per year.

Compare two different ways actors and writers on TV shows are paid for their work.

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1.02 Income tax

Worksheets
Income tax tables
Income tax and Medicare levy

Not all of a person's income is taxed. If we use some of our income for work-related expenses or to donate money to charities, these amounts are called **allowable deductions** (or tax deductions) and are not taxed. Examples of allowable deductions are tools of trade, uniforms, car/travel expenses, subscriptions to professional organisations and journals.

Income tax

Income tax is calculated on a person's **taxable income**, which is the gross income (total earnings) less all allowable deductions, rounded down to the nearest dollar.

Taxable income = gross income − allowable deductions

Video
Income tax

The more a person earns, the higher the rate of tax to be paid.

Tax rates for Australian residents	
Taxable income	**Tax on this income**
0 – $18 200	Nil
$18 201 – $45 000	19c for each $1 over $18 200
$45 001 – $120 000	$5092 plus 32.5c for each $1 over $45 000
$120 001 – $180 000	$29 467 plus 37c for each $1 over $120 000
$180 001 and over	$51 667 plus 45c for each $1 over $180 000

Source: © Australian Taxation Office for the Commonwealth of Australia

Example 6

Sophia earned $62 348 last financial year and collected bank interest of $440.81. She had allowable deductions of $427.52 in work expenses and $110 in donations to charities.

a Calculate her taxable income.

b Use the tax table to calculate the income tax that Sophie must pay.

Solution

a Taxable income = $62 348 + $440.81 − $427.52 − $110

= $62 251.29

≈ $62 251 Rounded down to the nearest dollar

b According to the table, a taxable income of $62 251 is in the $45 001 − $120 000 tax bracket.

Income tax = $5092 + 0.325 × ($62 251 − $45 000) '32.5c for each $1' means 32.5% or 0.325

= $10 698.575

≈ $10 698.58

PAYG tax and net pay

Income tax deducted from your pay by your employer every payday is called **PAYG (Pay As You Go) tax**. The total amount of PAYG tax paid over the year is usually more than the actual income tax payable, so at the end of the financial year you will receive the difference as a tax refund.

Gross pay is the total amount a person earns or receives, but most workers have a variety of deductions taken from their pay before they receive it, including PAYG tax, superannuation contributions, union fees and health fund payments. The amount of income left after the deductions is called **net pay**.

Net pay

Net pay = gross pay − tax − other deductions

Example 7

Jayden earns a gross pay of $2290.33 per fortnight. His deductions are for PAYG tax, $44.10 for private health insurance and $55.82 for superannuation.

Fortnightly earnings ($)	PAYG tax withheld ($)
2270–2275	418
2276–2279	420
2280–2285	422
2286–2291	424
2292–2297	426
2298–2303	428

a Use the PAYG tax table to find Jayden's PAYG tax per fortnight.

b Calculate Jayden's net pay.

c Calculate Jayden's total deductions as a percentage of his gross income (correct to one decimal place).

Solution

a In the table, $2290.33 falls in the $2286 – $2291 range.

Fortnightly PAYG tax = $424

b Net pay = $2290.33 – ($424 + $44.10 + $55.82)

= $2290.33 – $523.92

= $1766.41

Net pay = gross pay − total deductions

c Total deductions = $523.92

$$\text{Deductions percentage} = \frac{\$523.92}{\$2290.33} \times 100\%$$

$$= 22.8753...\%$$

$$\approx 22.9\%$$

$$\frac{\text{Total deductions}}{\text{Gross pay}} \times 100\%$$

EXERCISE 1.02 ANSWERS ON P. 576

Income tax U F PS C

EXAMPLE 6

1 Suemin earns $49 628 in a year and has allowable deductions of $1930.46.

a Calculate her taxable income, rounded down to the nearest dollar.

b Use the tax table on page 11 to calculate the income tax that Suemin must pay.

2 Aiden is an environmental engineer who had a gross income of $118 742 this year and work-related expenses totalling $4022.80, which are tax-deductible. Calculate Aiden's:

a taxable income, rounded down to the nearest dollar

b income tax.

3 Ellie is a graphic designer who earns an annual salary of $90 541 and has collected $1029.45 in bank interest. She has allowable deductions of $379 for tools related to her work and $287 in donations to charity. Calculate: PS

a Ellie's taxable income **b** the amount of tax payable.

4 Riley the builder had a gross income of $56 922 this year. He is entitled to these tax deductions: tools $1538, training courses $445 and outdoor protective clothing $506. How much should Riley pay in tax? Select the correct answer **A**, **B**, **C** or **D**. PS

A $9775.58 **B** $8966.65 **C** $8157.73 **D** $6884.27

5 Nicola is a nurse earning $87 996 per year. Her allowable deductions are the cost of non-slip footwear $225, the cost of laundering uniforms $1046, and union fees $297.60. How much should Nicola pay in tax? PS

6 Will owns a photography business and earned $228 000 last year. His allowable deductions were website costs $968, photographic equipment $23 672, and travel to photo shoots $15 930. Calculate the amount that Will should pay in tax. PS



Dean Drobot/Shutterstock.com

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7 Jackson earns a gross weekly income of $1075.26. His weekly deductions are $309.11 PAYG tax, $44.55 for private health insurance and $25.18 for superannuation. Calculate Jackson's net weekly pay. C

8 Elsa earns a gross income of $798.20 per week. Her deductions are $132.44 tax and $32.24 for private health insurance. Calculate Elsa's net income. C

Use the PAYG table from Example **7** on page 13 to answer questions **9** to **12**.

1.02

EXAMPLE 7

9 Every fortnight, Mr Singh earns $2278 and pays $22.80 in union fees and $94.10 in superannuation.

- **a** Find how much PAYG tax he pays per fortnight.
- **b** Calculate Mr Singh's fortnightly net pay.
- **c** What percentage (correct to one decimal place) of his gross pay do the deductions make up?

10 Holly earns a gross pay of $2295 per fortnight. Her deductions are PAYG tax, $64.35 for superannuation and $30 for life insurance. Find Holly's:

- **a** PAYG tax
- **b** net pay
- **c** total deductions as a percentage of her gross income (correct to one decimal place).

11 Stefan earns $1148 per week.

- **a** If he is paid fortnightly, what is his fortnightly gross pay?
- **b** Find the PAYG tax that is taken out of his gross pay.
- **c** Stefan's deductions are $141.94 for his health fund and $51.33 for superannuation. Calculate Stefan's net fortnightly pay.

12 Asmita earns a salary of $59 945 p.a. Each fortnight she has deductions of $256.20 for family health insurance and $35 for superannuation taken from her gross income.

- **a** Calculate Asmita's fortnightly gross income.
- **b** How much PAYG tax does she pay per fortnight?
- **c** Calculate Agata's fortnightly net income.

13 Copy and complete this pay slip. PS C

Employee: Ziad Chaker		Hourly pay rate: $21.65	
Hours worked		Deductions	
Normal	38	Tax: $205.72	Other: $168.38
Time-and-a-half	3	Gross weekly income	
Double time	2	Total deductions	
		Net weekly income	

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Technology

Online income tax calculators

The Australian Taxation Office (ATO) website **www.ato.gov.au** has online calculators for income tax and PAYG tax. Visit the website and search 'Simple Tax Calculator' to find the income tax calculator for individuals.

1. Enter the taxable income $63 000 as '63000' (no spaces).
2. Select the current financial year.
3. Select 'Resident for full year' and click 'Next'.
4. The estimated tax payable will be shown on a new screen.
5. Repeat for at least 2 more taxable incomes.
6. Find the PAYG tax calculator and use it to find the PAYG tax payable and net pay for a gross pay of:
   - a $1408 weekly
   - b $2870 fortnightly
   - c $5610 monthly

1.03 Simple interest

Worksheets
Simple interest
Simple interest table

Puzzle
Simple interest

Video
Simple interest

- When you invest money, you receive interest from your investment.
- When you borrow money, you pay interest on your loan.
- The original amount of money invested or borrowed is called the **principal**.
- This interest rate is a percentage of the principal, usually written as a rate **per annum** ('per year'), abbreviated 'p.a.'
- **Simple interest** (or flat rate interest) is interest calculated simply on the original principal.

The simple interest formula

$I = Prn$, where:

I is the simple interest

P is the principal

r is the interest rate per time period, expressed as a decimal

n is the number of time periods

Example 8

Find the simple interest on:

a \$4000 at 3.5% p.a. for 6 years

b \$13 500 at 5.5% p.a. for 7 months

c \$75 640 at 0.42% per month for 2 years

Solution

a $P = \$4000$, $r = 3.5\% = 0.035$, $n = 6$ years

$I = Prn$
$= \$4000 \times 0.035 \times 6$
$= \$840$

$r = 0.035$ per year, $n = 6$ years, so the time period is **years**

b $P = \$13\,500$, $r = 5.5\% = 0.055$, $r = \frac{7}{12}$ years

$I = Prn$
$= \$13\,500 \times 0.055 \times \frac{7}{12}$
$= \$433.125$
$\approx \$433.13$

$r = 0.055$ per year, $n = 7$ *months*, so we must change 7 months to years so that the time period is **years**

rounded to the nearest cent

c $P = \$75\,640$, $r = 0.42\% = 0.0042$,
$n = 2 \times 12 = 24$ months
$I = Prn$
$= \$75\,640 \times 0.0042 \times 24$
$= \$7624.512$
$\approx \$7624.51$

$r = 0.0042$ per month, $n = 2$ *years*, so we must change 2 years to months so that the time period is **months**

rounded to the nearest cent

Example 9

Petra invests \$17 400 for 2 years at 3.75% p.a. flat rate interest. To what final value will her investment grow?

Solution

$P = \$17\,400$, $r = 3.75\% = 0.0375$, $n = 2$ years

Both n and r are in **years**

$I = Prn$
$= \$17\,400 \times 0.0375 \times 2$
$= \$1305$

Value of investment $= \$17\,400 + \$1305$
$= \$18\,705$

Principal + interest

Video
Simple interest rate

Example 10

After 4 years, an investment of \$13 000 has earned \$1092 in simple interest.

What is the annual interest rate?

Solution

$I = \$1092$, $P = \$13\,000$, $n = 4$ years

$$I = Prn$$

$$\$1092 = \$13\,000 \times r \times 4$$

$$\$1092 = \$52\,000r$$

$$n = \frac{\$1092}{\$52000}$$

$$= 0.021$$

$$= 2.1\%$$

$\therefore$ Annual interest rate $= 2.1\%$

Video
Simple interest rate

Example 11

For how many months will \$10 000 need to be invested to earn \$250 in simple interest at 3.25% p.a.?

Solution

$I = \$250$, $P = \$10\,000$, $r = 3.25\% = 0.0325$

$$I = Prn$$

$$\$250 = \$10\,000 \times 0.0325 \times n$$

$$\$250 = \$325\text{n}$$

$$n = \frac{\$250}{\$325}$$

$= 0.7692...$ years — n is in years, so convert to months

$= 0.7692... \times 12$ months

$= 9.230...$ months

≈ 10 months — rounded *up* to the nearest month

EXERCISE 1.03 ANSWERS ON P. 576

Simple interest U F PS R

EXAMPLE 8

In this exercise, round all money answers to the nearest cent.

1 Calculate the simple interest earned on each investment.

- **a** \$35 000 for 4 years at 3.6% p.a.
- **b** \$26 850 at 1.95% p.a. for 2 years
- **c** \$8200 invested for 5 months at 3% p.a.
- **d** \$6590 invested for 16 months at 0.25% per month
- **e** \$5250 invested for 250 days at 1.04% p.a.
- **f** \$18 400 invested for 3 years at 0.18% per month

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2 Calculate the flat rate interest charged on each loan.

a $1250 for 2 years at 3.5% p.a.

b $18 900 for $5\frac{1}{2}$ years at 5.7% p.a.

c $1.15 million at 4.5% p.a. for 48 months

d $12 000 for 10 months at 0.575% per month

e $9750 for 2.5 years at 0.48% per month

f $24 720 for 136 days at 7.85% p.a.

3 Harry owed $783.26 on his credit card. The credit card company charged him one month's simple interest at 21% p.a. How much interest was he charged? Select the correct answer **A**, **B**, **C** or **D**.

A $13.71 **B** $16.45 **C** $25.38 **D** $37.30

EXAMPLE 9

4 Find the final value of each investment using simple interest.

a $10 000 invested for 3 years at 4% p.a.

b $1500 invested for 18 months at 0.19% per month

c $8500 invested for 3.5 years at 0.25% per month

d $9250 invested for 50 months at 3.75% p.a.

5 Liong borrowed $6000 to go on an overseas holiday, at 12% p.a. flat rate interest for 2 years. Calculate:

a the total interest

b the total amount Liong must repay.



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Foundation Standard Complex

EXAMPLE 10

6 The interest on a loan of $2500 over 4 years is $450. Calculate the flat rate of interest p.a. R

7 Katy took out a loan for $22 000 over 5 years. If her total loan repayments amounted to $28 400, calculate: PS R

a the interest charged

b the flat rate of interest p.a., correct to 2 decimal places.

8 After 5 years, the interest on a loan of $8000 amounts to $2340. Calculate the annual simple interest rate.

EXAMPLE 11

9 For how many years will $4200 need to be invested to earn $200 interest, if the interest rate is 2.5% p.a.? R

10 How many weeks will it take for $50 000 to earn $750 in interest if the rate is 2.6% p.a.? R

11 How many days will it take for $20 000 to earn $300 in interest if the rate is 4% p.a.? R

12 An online bank offered the following investment to its customers.

- A rate of 2.35% p.a. simple interest for the first 4 months only
- Then the principal and interest reinvested at 0.45% p.a. simple interest

What will Dariya's investment of $3480 be worth after 7 months?
Select **A**, **B**, **C** or **D**. PS R

A $31.21 **B** $374.10 **C** $3511.21 **D** $3854.10

13 For how many months will $20 000 need to be invested to amount to $22 000, if interest is paid at the rate of 0.33% per month? PS R

14 What is the flat rate of interest (as a percentage p.a., correct to one decimal place) when $1650 earns $85 in interest over 2 years? R

15 Toula used a credit card to buy a netbook computer for $799 and some extra accessories for $246. She pays off this debt in 30 days. The credit card charges 22% p.a. simple interest. PS R

a Calculate the simple interest charged.

b How much will Toula pay after 30 days?

16 Bhashine earned $185 interest each year for 4 years on an investment account. At the end of the 4 years, she closed her account and withdrew $6000 in total.
What was the annual flat rate of interest paid into Bhashine's account?
Select **A**, **B**, **C** or **D**. PS R

A 3.1% **B** 3.5% **C** 12.3% **D** 14.1%

Foundation Standard Complex

Compound interest

1.04

Most investments earn **compound interest** rather than simple interest. With compound interest, the interest earned is **added** to the principal so that next time, the interest is calculated on a larger principal. This means that more interest is earned, because we are also earning interest on the interest we have already earned. The word **compound** means 'combined'.

Example 12

A principal of $23 000 is invested at 4% p.a. interest, compounded yearly for 2 years.

a What is the final value of the investment after 2 years?

b What is the amount of compound interest earned?

Solution

a The interest for each year is calculated separately.

After the first year:

$I = \$23\,000 \times 0.04$

$= \$920$

$\text{Investment} = \$23\,000 + \$920$ Principal + interest

$= \$23\,920$

After the second year:

$I = \$23\,920 \times 0.04$

$= \$956.80$

$\text{Investment} = \$23\,920 + \$956.80$ New principal + interest

$= \$24\,876.80$

b $\text{Compound interest earned} = \text{final value} - \text{principal}$

$= \$24\,876.80 - \$23\,000$

$= \$1876.80$

Notice that compound interest involves repeated percentage increase. In the above example, to calculate compound interest on a principal of $23 000 over 2 years at 4% p.a., we are actually increasing $23 000 by 4% twice.

Adding 4% to the principal is the same as increasing the principal by 4%, which is the same as multiplying the principal by 104% or 1.04.

Investment after 1st year $= \$23\,000 \times 1.04 = \$23\,920$

Investment after 2nd year $= \$23\,920 \times 1.04 = \$24\,876.80$

We can even combine these 2 steps into one step by repeated percentage increases:

Investment after 2nd year $= \$23\,000 \times 1.04 \times 1.04 = \$24\,876.80$

Using repeated percentage increases can simplify our compound interest calculations.

Example 13

A principal of $9000 is invested at 3.7% p.a. compounded yearly over 3 years. What is:

a the final value of the investment after 3 years?

b the compound interest earned?

Solution

a Adding 3.7% interest to the principal is the same as multiplying the principal by 1.037.

$\therefore$ final value after 3 years $= \$9000 \times 1.037 \times 1.037 \times 1.037$

$= \$9000 \times (1.037)^3$

$= \$10\,036.4188...$

$\approx \$10\,036.42$ rounded to the nearest cent

b Compound interest earned = final value – original principal

$= \$10\,036.42 - \$9000$

$= \$1036.42$

EXERCISE 1.04 ANSWERS ON P. 576

Compound interest U F

In this exercise, round all money answers to the nearest cent.

EXAMPLE 12

1 A principal of $23 000 is invested at 5% p.a. interest, compounded yearly over 2 years.

a Copy and complete the following working to calculate the value of the investment after 2 years.

After the first year:

$I = \$23\,000 \times 0.05$ Investment $= \$23\,000 + \$_____$

$= \$_____$ $= \$_____$

After the second year:

$I = \$_____ \times 0.05$ Investment $= \$_____ + _____$

$= \$_____$ $= \$_____$

b Copy and complete the following working to calculate the amount of compound interest earned.

Compound interest earned = final value – principal

$= \$_____ - \$23\,000$

$= \$_____$

2 Finn invests $15 000 at 2.5% p.a. compounded yearly over 3 years. Show all working (as in question **1**) to find:

a the final value of the investment after 3 years

b the total amount of compound interest earned.

3 Selina invests $34 100 at 6.2% p.a. interest compounded yearly over 2 years. Calculate:

a the final value of the investment **b** the compound interest earned.

4 Use repeated percentage increases to calculate the final value of each investment compounded annually, then calculate the compound interest earned.

a $5000 for 2 years at 4% p.a. **b** $27 800 for 3 years at 2.85% p.a.

c $9600 for 3 years at 5% p.a. **d** $39 500 for 2 years at 3% p.a.

e $18 400 for 4 years at 1.25% p.a.

5 For each investment, calculate the compound interest earned.

a $30 400 at 5% p.a. interest for 3 years.

b $19 150 at 4.2% p.a. interest for 2 years.

c $8750 at 1.75% p.a. interest for 2 years.

d $36 000 at 3.5% p.a. interest for 3 years.

e $18 960 at 6.35% p.a. interest for 5 years.

EXAMPLE 13

1.04

Mental skills 1: Maths without calculators ANSWERS ON P. 576

Quiz
Mental skills 1

Percentage of a quantity

Learn these commonly-used percentages and their fraction equivalents.

Percentage	50%	25%	12.5%	75%	20%	10%	$33\frac{1}{3}\%$	$66\frac{2}{3}\%$
Fraction	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{3}$	$\frac{2}{3}$

Now we will use them to find a percentage of a quantity.

1 Study each example.

a $20\% \times 25 = \frac{1}{5} \times 25$
$= 5$

b $50\% \times 120 = \frac{1}{2} \times 120$
$= 60$

c $12.5\% \times 32 = \frac{1}{8} \times 32$
$= 4$

d $75\% \times 56 = \frac{3}{4} \times 60$
$= \left(\frac{1}{4} \times 60\right) \times 3$
$= 15 \times 3$
$= 45$

e $33\frac{1}{3}\% \times 27 = \frac{1}{3} \times 27$
$= 9$

f $66\frac{2}{3}\% \times 60 = \frac{2}{3} \times 60$
$= \left(\frac{1}{3} \times 60\right) \times 2$
$= 20 \times 2$
$= 40$

2 Now simplify each expression.

a $25\% \times 44$ **b** $33\frac{1}{3}\% \times 120$ **c** $20\% \times 35$ **d** $66\frac{2}{3}\% \times 36$

e $10\% \times 230$ **f** $12\frac{1}{2}\% \times 48$ **g** $50\% \times 86$ **h** $20\% \times 400$

i $75\% \times 24$ **j** $33\frac{1}{3}\% \times 45$ **k** $25\% \times 160$ **l** $10\% \times 650$

m $12.5\% \times 88$ **n** $66\frac{2}{3}\% \times 21$ **o** $20\% \times 60$ **p** $75\% \times 180$

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1.05 The compound interest formula



Worksheet
Compound interest

Puzzles
Compound interest with annual rests

Compound interest with non-annual rests

Technologies
Comparing interest rates

Simple and compound interest calculator

Spreadsheets
Comparing interest rates

Simple and compound interest



Videos
The compound interest formula

Could you owe more than America?

There is a formula for calculating the final amount of an investment earning compound interest. Note the following pattern.

- Final value of \$23 000 at 4% p.a. interest for 2 years = $\$23\,000 \times (1.04)^2$
- Final value of \$9000 at 3.7% p.a. interest for 3 years = $\$9000 \times (1.037)^3$
- Final value of \$18 960 at 6.35% p.a. interest for 5 years = $\$18\,960 \times (1.0635)^5$

Compound interest formula

$FV = PV(1 + r)^n$, where:

FV is the final value **or** **future value** of the investment

PV is the principal, also called the **present value**

r is the interest rate per compounding period, expressed as a decimal

n is the number of compounding periods

The compound interest is then calculated using this formula:

Compound interest = future value − present value

$$I = FV - PV$$

Example 14

For each investment, calculate:

i the future value of the investment

ii the compound interest earned if interest is compounded annually

a \$26 750 is invested at 4% p.a. for 3 years

b \$52 000 is invested at 3.8% p.a. for 5 years

Solution

a i $PV = \$26\,750$, $r = 4\% = 0.04$, $n = 3$

$FV = PV(1 + r)^n$

$= \$26\,750(1 + 0.04)^3$

$= \$26\,750(1.04)^3$

$= \$30\,090.112...$

$\approx \$30\,090.11$

The final value of the investment is \$30 090.11.

ii Compound interest = \$30 090.11 − \$26 750 $I = FV - PV$

= \$3340.11

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b **i** $PV = \$52\,000, r = 3.8\% = 0.038, n = 5$

$FV = PV(1 + r)^n$

$= \$52\,000(1 + 0.038)^5$

$= \$52\,000(1.038)^5$

$= \$62\,659.9597...$

$\approx \$62\,659.96$

ii Compound interest $= \$62\,659.96 - \$52\,000$

$= \$10\,659.96$

Example 15

Calculate the compound interest when $24 500 is invested at 6.3% p.a. for 5 years:

a compounded annually

b compounded monthly.



Video
Compound interest

Solution

a $PV = \$24\,500, r = 0.063, n = 5$

$FV = \$24\,500\,(1 + 0.063)^5$

$= \$24\,500\,(1.063)^5$

$= \$33\,253.1205...$

$\approx \$33\,253.12$

$I = \$33\,253.12 - \$24\,500$

$= \$8753.12$

b Because interest is compounded monthly, r and n must be expressed in months, not years.

$PV = \$24\,500, r = \frac{0.063}{12} = 0.00525$ per month, $n = 5 \times 12 = 60$ months

$FV = \$24\,500(1 + 0.005\,25)^{60}$

$= \$24\,500(1.005\,25)^{60}$

$= \$33\,543.701\,98$

$\approx \$33\,543.70$

$I = \$33\,543.70 - \$24\,500$

$= \$9043.70$

Note: More interest is earned when it is compounded monthly rather than yearly.

Why do you think this is so?

EXERCISE 1.05 ANSWERS ON P. 576

The compound interest formula U F PS R

In this exercise, round all money answers to the nearest cent.

1 An amount of $13 000 is invested at 5% p.a. interest, compounded annually over 2 years. Which expression represents the future value of the investment?
Select the correct answer **A**, **B**, **C** or **D**.

A $13\,000 \times 0.05 \times 2$ **B** $13\,000\,(1 + 0.05)^2$

C $13\,000 \times (0.05)^2$ **D** $13\,000\,(1 - 0.05)^2$

EXAMPLE 14

2 For each investment, where interest is compounded yearly, calculate:

i the final amount of the investment, A **ii** the compound interest, I, earned.

a $6500 invested at 7% p.a. for 6 years **b** $10 000 invested at 8.5% p.a. for 4 years

c $12 240 invested at 1.6% p.a. for 2 years **d** $34 600 invested at 4.9% p.a. for 5 years

e $8000 invested at 1.75% p.a. for 3 years

3 Calculate the amount of interest earned on an investment of $6500 if it is invested at 2.5% p.a. compounded annually for 8 years. Select **A**, **B**, **C** or **D**.

A $131.14 **B** $832.81 **C** $1300 **D** $1419.62

4 Find the amount of interest earned on one million dollars invested at 14.9% p.a. compounded annually for 6 years.

5 Find the amount of interest charged on a loan of $25 000 if it is borrowed over 10 years at 8% p.a. compounded annually. Select **A**, **B**, **C** or **D**.

A $31 250 **B** $28 973.12 **C** $28 589.72 **D** $20 000

6 Yasmin was given $2000 when she turned 3 years old. Her parents invested it at a 2% p.a. compounded annually. No deposits or withdrawals were made. Which expression can be used to determine how much money Yasmin had in the account when she turned 16? Select **A**, **B**, **C** or **D**. R

A $2000\,(1 + 0.02)^{13}$ **B** $2000\,(1 + 0.2)^{13}$

C $2000\,(1 + 0.02)^{16}$ **D** $2000\,(1 + 0.2)^{16}$

EXAMPLE 15

7 For each investment, calculate:

i the future value **ii** the interest earned.

a $10 000 for 5 years at 2.4% p.a., compounded monthly

b $35 500 for 10 years at 2% per half-year, compounded half-yearly

Half-yearly means 'twice a year' or 'every 6 months'

c $8900 for 2 years at 3% p.a., compounded quarterly

Quarterly means '4 times per year' or 'every 3 months'

d $42 000 for 5 years at 0.225% per month, compounded monthly

e $16 500 for 3 years at 2.6% p.a., compounded half-yearly

f $4900 for 1 year at 0.005% per day, compounded daily



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8 Find the total value of an investment of $4300 over 5 years at 4.6% p.a. interest, compounded every 6 months. Select the correct answer **A**, **B**, **C** or **D**.

A $4817.78 **B** $5384.27 **C** $5397.90 **D** $8506.24

9 **a** Reese invested $6000 for 2 years at a flat rate of 5% p.a. Calculate the amount of interest earned.

b Tegan invested $6000 for 2 years at an interest rate of 5% p.a. compounded annually. Calculate the amount of interest earned.

c Whose investment earned more interest? How much more?

10 A principal of $5000 is invested at 3.5% p.a. for 3 years. Match each compounding period for this investment to its correct expression for the future value of the expression.

a compounded yearly

b compounded half-yearly

c compounded quarterly

d compounded monthly

A $5000\left(1+\frac{0.035}{2}\right)^{3\times2}$

B $5000\left(1+\frac{0.035}{12}\right)^{3\times12}$

C $5000\left(1+\frac{0.035}{4}\right)^{3\times4}$

D $5000\,(1+0.035)^3$

11 Lisa is setting up a trust account for her new grandson Stefan. In 18 years time, she wants the investment to be worth $30 000, to help with the cost of university fees or the purchase of a car. Suppose the interest rate for the account is 2.94% p.a. compounding yearly. PS R

a How much should Lisa invest now to achieve the $30 000 target?

b If Lisa opened a trust account that earns 2.94% p.a. compounding monthly instead, how much less would she need to invest?

12 Zoe is 5 years old and about to start school. Her parents want to invest $25 000, for her high school education expenses, in an account that earns 3% p.a. over 7 years. R

a Calculate the total interest earned if interest is compounded:

i yearly **ii** half-yearly **iii** quarterly **iv** monthly

b Which compounding period should Zoe's parents choose? Why?

13 A principal of $10 000 is invested for 4 years, earning interest at the rate of 3% p.a., compounded monthly. Which expression represents the future value of the investment? Select **A**, **B**, **C** or **D**.

A $10\,000\left(1+\frac{3}{100}\right)^4$

B $10\,000\left(1+\frac{3}{100}\right)^{48}$

C $10\,000\times\left(1+\frac{3}{1200}\right)^4$

D $10\,000\left(1+\frac{3}{1200}\right)^{48}$

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Skillsheet
Spreadsheets

Technology

Comparing simple with compound interest

In this activity, you will compare the interest earned on an investment of $1000 for 10 years at 8% p.a. simple interest and 8% p.a. compound interest, compounded annually.

1 Create this spreadsheet. The present value (*PV*) is entered in cell A1 and the annual interest rates (in decimal form) in cells B1 and C1.

	A	B	C
1	$ 1,000.00	0.08	0.08
2			
3	**Years**	**Simple Interest**	**Compound Interest**
4	1		
5	2		
6	3		
7	4		
8	5		
9	6		
10	7		
11	8		
12	9		
13	10		

2 To calculate the simple interest in column B, in cell B4 enter the formula **=$A$1*$B$1*A4**. Now **Fill Down** from cell B4 to B13.

3 To calculate the compound interest in column C, in cell C4 enter the formula **=$A$1*(1+$C$1)^A4-$A$1**. Now **Fill Down** from cell C4 to C13.

4 Highlight cells A3 to C13. **Insert 'Scatter with Smooth lines and markers'**.

5 When the interest rate is the same, which account pays better interest: simple or compound interest? (Type your answer in cell A15)

6 Now compare the interest earned on an investment of $1000 for 10 years at 9% p.a. simple interest and 7% p.a. compound interest, compounded annually. Change the interest rates in cells B1 (0.09) and C1 (0.07) respectively.

Answer the following questions in the spreadsheet cells indicated in brackets.

7 After how many years did the compound interest rate pay more than the simple interest rate? (A16)

8 How much extra interest did the compound interest rate pay at the end of the 10 years? (A17)

9 Change the interest rate in B1 to 10% (0.1) and C1 to 9% (0.09). How does the change in interest rate affect the amount of interest paid? Include calculations to justify your answer. (A18)

10 Change the interest rate in B1 to 12% (0.12) and C1 to 8.5% (0.085). After how many years did the amount of compound interest earned overtake the amount of simple interest earned? (A19)

11 What is the difference in the amount of compound interest earned for the 10-year period compared to the simple interest investment? Is it a significant amount? Justify your answer. (A20)

Term payments

1.06

Many customers buy expensive household items **on terms**, which means 'paying off' the item by regular instalments over time, after paying a deposit. A **term payments** plan is also called **hire-purchase** because the purchaser actually hires the item until it is completely paid off. Special offers can include interest-free periods, but there may be other conditions such as establishment fees and extra charges if the regular **repayments** are not paid on time. Also, if the purchaser fails to keep up with the payments, higher interest may be charged or the item may be **repossessed** (taken back).



Technology
Term payments

Example 16

Sonia purchases a new fridge and dishwasher package valued at $4925. She pays a 10% deposit and repays the balance in monthly repayments over 3 years. Interest on the balance is charged at a flat rate of 12% p.a.

Find:

a the deposit paid

b the balance owing

c the interest charged on the balance

d the total to be repaid

e the amount of each monthly repayment

f the total price paid for the package.

Solution

a Deposit $= 10\% \times \$4925$

$= \$492.50$

b Balance owing $= \$4925 - \$492.50$ or $90\% \times \$4925$

$= \$4432.50$

c Interest charged on the balance is flat or simple interest.

$P = \$4432.50$, $r = 0.12$, $n = 3$

$I = Prn$

$= \$4432.50 \times 0.12 \times 3$

$= \$1595.70$

d Total to be repaid = balance + interest

$= \$4432.50 + \$1595.70$

$= \$6028.20$

e Number of repayments $= 3 \times 12$ months

$= 36$ months

Monthly repayment $= \$6028.20 \div 36$

$= \$167.45$

f Total price paid = deposit + total repayments

$= \$492.50 + \$6028.20$

$= \$6520.70$

or cash price + interest

$= \$4925 + \$1595.70$

Deferred payment plan

With a **deferred payment plan**, the customer does not make any repayments until a later date, such as after 6 months.

Deferred means 'delayed'.

Example 17



Viktorus/Shutterstock.com

Ilhea and Robert purchase a $3200 home theatre package on a deferred payment plan over 2 years. They make no repayments for the first 3 months, then pay $210 per month.

a Calculate the total repayments.

b How much interest did they pay?

c Ilhea and Robert were also charged the following fees for the plan.

- One-off establishment fee of $25
- Account service fee of $9.95 per month

i How much was paid in fees over the 2-year period?

ii What percentage (correct to one decimal place) of the purchase price was paid in fees?

Solution

a Total cost $= \$210 \times 21$ 2 years − 3 months = 21 months

$= \$4410$

b Interest = total cost − cash price

$= \$4410 - \$3200$

$= \$1210$

c **i** Total fees paid $= \$25 + \$9.95 \times 24$

$= \$263.80$

ii Percentage $= \frac{\$263.80}{\$3200} \times 100\%$

$= 8.24375\%$

$\approx 8.2\%$

EXERCISE 1.06 ANSWERS ON P. 577

Term payments U F PS R C

In this exercise, round all money answers to the nearest cent.

EXAMPLE 16

1.06

1 Gaspard purchases a laundry package of washing machine and dryer valued at $1755. He pays a 10% deposit and repays the balance in 18 months. Interest on the balance is charged at a flat rate of 13.85% p.a. Find: PS R C

- **a** the deposit Gaspard paid
- **b** the balance owing
- **c** the interest charged on the balance
- **d** the total to be repaid
- **e** the amount of each monthly instalment
- **f** the total price paid for the package

2 Freyja bought a backyard plunge pool for $25 500. She paid a 5% deposit and will repay the balance over 5 years, with flat-rate interest charged 8.6% p.a. Calculate: PS R C

- **a** the deposit
- **b** the balance owing
- **c** the interest charged
- **d** the total to be repaid
- **e** the monthly instalment
- **f** the total price paid.

3 Tracy wants to buy a new 3-piece lounge suite that she saw advertised at the lounge sale pictured, costing $6895. R



- **a** How much deposit will she have to pay?
- **b** What will be the flat-rate interest on the balance?
- **c** Find the size of each monthly repayment.

4 Mack pays a $1200 deposit on his holiday and then $185 per month for a year. The cash price of the holiday is $2700.

- **a** How much did Mack pay for his holiday?
- **b** Calculate the interest paid.
- **c** Calculate the balance owing after the deposit was paid.
- **d** Calculate, as a percentage correct to one decimal place, the flat interest rate charged p.a.

5 Mrs Mitchell buys a used car for her daughter by paying a 20% deposit and $275 per month for 4 years. If the car has a cash price of $10 400, find: R C

- **a** the deposit Mrs Mitchell paid
- **b** the balance owing after the deposit is paid
- **c** the total amount Mrs Mitchell repaid in instalments
- **d** the interest charged
- **e** the flat interest rate (p.a.), correct to one decimal place.

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6 Xuan wanted to buy a new smartphone costing $1044. R C

a How much deposit did she pay?

b What is the total amount repaid in repayments for the smartphone?

c Calculate the total interest paid for the year.

d What was the annual flat interest rate for this purchase, correct to 1 decimal place?



Cinclia/Shutterstock.com

EXAMPLE 17

7 Diego bought a laptop priced at $1500 on a deferred payment plan: nothing to pay for 3 months and then 9 monthly payments of $210. However, a monthly account fee of $12.95 was added to the plan. PS R

a Find the total cost of the laptop to Diego.

b How much in excess of the cash price was paid?

c What was the flat rate of interest p.a. (correct to one decimal place) charged?

'in excess' means 'above' or 'more than'

8 Tahlia bought a new outdoor setting and BBQ for her backyard that retails for $3899. She paid $500 deposit, no payments for 6 months and then fortnightly payments of $78 for 2 years. PS R

a Find the balance owing after Tahlia had paid the deposit.

b Calculate the total cost of the outdoor setting and BBQ.

c How much in excess of the cash price did Tahlia pay under this plan?

d What is the annual flat rate of interest charged, correct to one decimal place?

9 Sofia bought a home cinema system priced at $2100 on interest-free terms for one year with no repayments for the first 4 months. R

a If Sofia makes 8 equal monthly repayments, what is the amount of each payment?

b There is a service charge of $12.95 every month for this deferred payment plan. What percentage (correct to one decimal place) of the purchase price was paid in service charges?

Investigation

Short-term loans: fast money

Perhaps you have seen ads from finance companies (not banks) offering to lend you cash for a short amount of time (from 16 days to 12 months) if you can't afford something, with quick (or even instant) approval regardless of your financial situation. These small loans are meant to 'tie you over' if you are short of money, but there is always a catch.

In this activity, you will compare the borrowing costs associated with small amounts borrowed as short-term loans.

1 Alice takes out a short-term loan of $700 with an interest rate of 18% per week.

- **a** If she repays the loan in 2 weeks, calculate the total amount Alice will need to repay.
- **b** What do you notice about the amount of interest charged?
- **c** What do you think will happen if Alice cannot repay the loan in 2 weeks?

2 Disha borrows $1500 for one month and pays a flat fee of $30 per $100 borrowed.

- **a** Calculate the total cost of the loan to be repaid.
- **b** What do you notice about the amount of interest charged?
- **c** Suggest a reason a person may need to borrow $1500 for a short time but cannot use a credit card or bank loan.

3

- **a** Do an Internet search of 'short-term loans' and find 2 Australian finance companies that provide short-term loans.
- **b** In addition to charging interest, short-term loans have a range of different fees and costs. List the fees and costs charged by the 2 finance companies that you found.
- **c** List the interest rates of some short-term loans provided by the finance companies.
- **d** List 2 advantages and 2 disadvantages of taking a short-term loan.

4 For a short-term loan of $1500, consider weekly, fortnightly and monthly payments for each of the 2 companies. Copy and complete this table.

Repayments and fees	Company 1	Company 2
Weekly		
Fortnightly		
Monthly		

5 Repeat Question 4 for a short-term loan of $2000.

6 What would be your advice to someone considering a short-term loan? Analyse and compare the loan calculations from the 2 companies you have chosen to support your observations.

1.07 Depreciation

Worksheet
Depreciation

Depreciation is the decrease in value of an item over time, usually something expensive and valuable, called an **asset**. When items lose value because of age or frequency of use, they are said to **depreciate**.

The compound interest formula can be adapted to find the depreciated value of an asset. While compound interest involves repeated percentage increases, depreciation involves repeated percentage *decreases*, so its formula has a minus sign.

Depreciation formula

$S = V_0(1 - r)^n$, where:

S is the final value of the asset, called the **salvage value**

V_0 is the **initial (original) value** of the asset (the value V at time 0)

r is the rate of depreciation per period, expressed as a decimal

n is the number of periods of depreciation

The amount of depreciation is then calculated using this formula:

$$\text{Depreciation} = \text{original value} - \text{final value}$$

Video
Depreciation

Example 18

An accountant's computer system depreciates by 15% each year.

a If the computer system is currently valued at \$2600, what will its value be in 5 years?

b What is the depreciation over this time?

Solution

a $V_0 = \$2600$, $r = 15\% = 0.15$, $n = 5$

$S = V_0(1 - r)^n$

$= \$2600\ (1 - 0.15)^5$

$= \$2600\ (0.85)^5$

$= \$1153.6338...$

$\approx \$1153.63$

The value of the computer system in 5 years will be \$1153.63.

b Depreciation $= \$2600 - \$1153.63$ original value − final value

$= \$1446.37$

9780170479622

Example 19

An industrial oven in a restaurant originally costs $19 800, then depreciates at a rate of 12% p.a.

a Find the salvage value of the oven after 6 years, correct to the nearest dollar.

b Express the salvage value as a percentage of the cost price, correct to one decimal place.

Solution

a $V_0 = \$19\,800$, $r = 0.12$, $n = 6$

$S = V_0(1 - r)^n$

$= \$19\,800\,(1 - 0.12)^6$

$= \$19\,800\,(0.88)^6$

$= \$9195.2009...$

$\approx \$9195$

b Percentage of cost price $= \frac{\$9195}{\$19\,800} \times 100\%$

$= 46.4393...\ \%$

$\approx 46.4\%$

This means that after 6 years, the oven is worth approximately 46% of its original price (or has lost 54% of its initial value).



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EXERCISE 1.07 ANSWERS ON P. 577

Depreciation U F PS R C

In this exercise, round all money answers to the nearest cent.

EXAMPLE 18

1 Find the salvage value of a photocopier after 5 years if its purchase price was $2850 and the annual depreciation rate is 20%.

2 **a** Find the value of a car after 7 years if it is purchased new for $49 990 and it depreciates at 12% p.a.

b Find the amount of depreciation over this time.

3 For each item shown in the table, calculate:

i its value after 4 years of depreciation

ii its value after 4 years as a percentage of its original value, correct to one decimal place.

	Item	Original value	Depreciation rate (p.a.)
a	Oven	$1100	12%
b	Fishing boat	$38 500	18%
c	Library	$8460	12%
d	Computer	$2500	20%
e	Furniture	$27 500	15.5%
f	Bike	$2900	22%
g	Electrical tools	$870	17.5%
h	Air conditioner	$2600	9%

EXAMPLE 19

4 A smartphone originally valued at $1729 depreciates at 37% p.a. R

a What percentage (to 2 decimal places where necessary) of the original value remains after:

i 1 year? **ii** 3 years? **iii** 6 years?

b Approximately how long would it take the smartphone to halve its original value?

5 A security system costs a company $12 500 to buy new. It depreciates at a yearly rate of 20%.

a Find the salvage value of the system after:

i 1 year **ii** 2 years **iii** 5 years

b Find the value of the system after 5 years as a percentage of its initial value. Answer correct to one decimal place.



6 Parveen pays $25 490 for a new car. The car will depreciate in value by an average of 11% p.a.

- **a** Find correct to the nearest dollar the market value of the car in 3 years.
- **b** Calculate the amount of depreciation in the car after 3 years.

7 Adam has spent $175 000 on equipment to set up his hairdressing salon. The equipment depreciates at 20% per year. R

- **a** Find the value of the equipment after 4 years.
- **b** Find the amount of depreciation in the equipment after 4 years.
- **c** Find, by trial and error, how long it will take for the value to be under $50 000. Answer in years and months.
- **d** Find the value of Adam's equipment after 9 years as a percentage of its original value, correct to one decimal place.

8 Kamal says that, at 10% p.a. depreciation, a car will lose half its value after 7 years.

Is he correct? Show all working to justify your answer. PS R C

9 Office equipment that is worth $12 000 when new, depreciates at 15% p.a. as shown in the table. PS R C

- **a** How much did the office equipment lose in value in the first year?
- **b** After how many years did the office equipment fall below half its original value?
- **c** By how much did the office equipment depreciate between the 5th and 6th years?
- **d** Will the value of the office equipment ever fall below $100?
- **e** Will the value of the office equipment ever be zero?

Year	Depreciated value
0	$12 000
1	$10 200
2	$8670
3	$7369.50
4	$6264.08
5	$5324.46
6	$4525.79
7	$3846.93
8	$3269.89
9	$2779.40
10	$2362.49
11	$2008.12
12	$1706.90
13	$1450.87
14	$1233.24

Power plus ANSWERS ON P. 577

1 How long, in years and days, will it take an investment of $3000 to earn $500 in simple interest at 4% p.a.?

2 What amount should Owen invest to earn $100 in simple interest if the investment will last for 9 months and the interest rate is 3% p.a.?

3 A principal of $10 000 is invested for 5 years at an interest rate of 5% p.a., with interest compounded weekly. Calculate the future value of the investment.

4 Meghan needs $80 000 in 4 years' time. What present value should she invest now at an interest rate of 6% p.a., with interest compounded annually, to reach her target?

5 A painting appreciates in value at a rate of 3% p.a. while a computer depreciates in value at a rate of 10% p.a. If I bought the painting for $1200 and the computer for $1500 new, what would be their combined value in 5 years time?

6 A bacteria colony is growing at a rate of 20% per hour. If there are 10 000 bacteria now, use the compound interest formula to calculate how many there will be after 1 day. (Give your answer correct to the nearest 10 000.)

7 a You invest $2000 in a bank account at an interest rate of 4% p.a. with interest compounded annually. How long will it take for your investment to double in value?

b If you invested $4000 instead of $2000 at the same interest rate, how long will it take to double in value?

c Does the size of the principal make any difference to the time taken for it to double?

CHAPTER 1 REVIEW

Language of maths

allowable deductions	annual leave loading	compound interest	deposit
depreciation	double time	flat rate	fortnightly
future value	income tax	initial value	overtime
PAYG tax	per annum (p.a.)	present value	principal
quarterly	repayment	salvage value	short-term loan
simple interest	taxable income	term payments	time-and-a-half

1 When investing, why is compound interest better than simple interest?

2 What do the r and n stand for in the formulas $I = Prn$ and $FV = PV(1 + r)^n$?

3 What is another name for flat-rate interest?

4 Which word means a decrease in the value of an item over time?

5 Explain the meanings of the terms **present value** and **future value**.

6 Use a dictionary to find at least 2 different meanings of principal.

Quiz
Language of maths 1

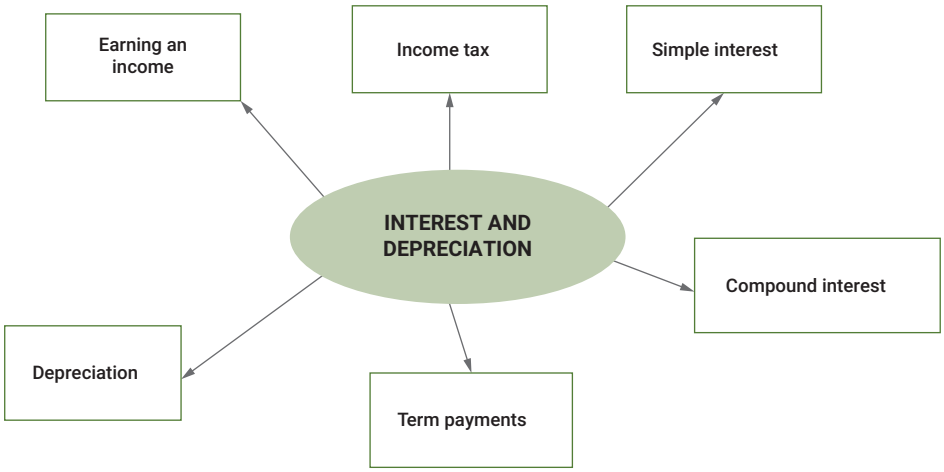
Puzzle
Interest and depreciation crossword

Topic summary

- Which parts of this chapter were revision of Year 9 knowledge and skills?
- Which parts of this chapter were new to you?
- Do you know how to use the simple interest and compound interest formulas?
- How is income tax calculated?
- How is the depreciation formula similar to the compound interest formula?

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Interest and depreciation



Quiz
Test yourself 1

TEST YOURSELF 1

ANSWERS ON P. 577

In this exercise, round all money answers to the nearest cent.

1.01 **1** Alexis is paid a commission of 2.5% on the value of the properties she sells. She also receives a weekly retainer of $1150. How much will Alexis earn if she sells a house for $475 830?

1.01 **2** Caleb earns a salary of $70 400 p.a. How much is he paid each week?

1.01 **3** A supermarket cashier is employed under the following award.

Normal rate: $21.45 per hour	
Normal rate	For 0 to 38 hours worked
Time-and-a-half	For the next 4 hours worked
Double time	For each hour worked after that

Calculate the wage for working:

a 40 hours

b 46 hours.

1.01 **4** Silvia earns royalties from writing cookbooks, at 15.5% of the price of the book.

Christmas sales figures of her latest book are shown in this table.

Version	Retail price	Sales
ebook	$25.95	8415
Hardcover	$45.95	26 500
Softcover	$35.95	15 260

How much did Silvia earn in royalties for these sales, rounded to the nearest cent?

1.01 **5** For his Christmas holidays, Jacan received 4 weeks normal pay plus 17.5% annual leave loading for the 4 weeks. If Jacan's annual salary is $54 920, find:

a his normal weekly pay

b his leave loading

c his total pay for the 4-week holiday.

1.02 **6** Alia earns a salary of $68 650 p.a. Her allowable deductions are donations to charities of $540 and work-related expenses of $385.

a Calculate Alia's taxable income.

b Use the tax table on page 12 to calculate the income tax Alia should pay.

1.03 **7** Calculate the simple interest earned on each investment.

a $20 000 invested for 3 years at 4% p.a.

b $7850 invested at 2.5% p.a. for 15 months

c $4500 invested for 6 months at 0.17% per month

d $25 200 invested for 100 days at 3.45% p.a.

8 An amount of $5000 is invested at 2.35% p.a. interest, compounded annually over 3 years. 1.04

a What is the total value of the investment after 3 years?

b What is the amount of compound interest earned?

9 Calculate the value of the investment when $34 200 is invested at 3% p.a. for 2 years, with interest compounded annually. 1.05

10 Find the final value if $11 000 is invested for 4 years at 2.4% p.a., with interest compounded monthly. 1.05

11 Find the interest earned when $4895 is invested at 1.95% p.a. for 3 years, with interest compounded annually. 1.05

12 Calculate the interest earned when $46 230 is invested for 9 years at 2.8% p.a., with interest compounded half-yearly. 1.05

13 Yang purchases a furniture package valued at $4875. She pays a 10% deposit and repays the balance in 36 monthly instalments. Interest on the balance is charged at a flat rate of 14.5% p.a. Find: 1.06

a the deposit Yang paid

b the balance owing

c the interest charged

d the total to be repaid

e the amount of each instalment

f the total price Yang paid for the package.

14 Bridget bought a new car for $24 990, which depreciates by 10% p.a. 1.07

a Find correct to the nearest dollar the depreciated value of the car after 5 years.

b What is the depreciation over this time?

c Express the depreciated value as a percentage of the original price (correct to one decimal place).